RingMo-SAM: A Foundation Model for Segment Anything in Multimodal Remote-Sensing Images

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***Abstract*— The proposal of the segment anything model (SAM) has created a new paradigm for the deep-learning-based semantic segmentation field and has shown amazing generalization performance. However, we find it may fail or perform poorly on multimodal remote-sensing scenarios, especially synthetic aperture radar (SAR) images. Besides, SAM does not provide category information for objects. In this article, we propose a foundation model for multimodal remote-sensing image segmentation called RingMo-SAM, which can not only segment anything in optical and SAR remote-sensing data, but also identify object categories. First, a large-scale dataset containing millions of segmentation instances is constructed by collecting multiple open-source datasets in this field to train the model. Then, by constructing an instance-type and terrain-type category-decoupling mask decoder (CDMDecoder), the categorywise segmentation of various objects is achieved. In addition, a prompt encoder embedded with the characteristics of multimodal remote-sensing data is designed. It not only supports multibox prompts to improve the segmentation accuracy of multiobjects in complicated remote-sensing scenes, but also supports SAR characteristics prompts to improve the segmentation performance on SAR images. Extensive experimental results on several datasets including iSAID, ISPRS Vaihingen, ISPRS Potsdam, AIR-PolSAR-Seg, and so on have demonstrated the effectiveness of our method.**

***Index Terms*— Multimodal remote-sensing images, prompt learning, segment anything model (SAM), semantic segmentation.**

I. Introduction

SEMANTIC segmentation is a fundamental and important task in the remote-sensing image interpretation field. It aims to extract and classify the whole images at the pixel level and has been widely used in various practical applications, such as urban planning, road extraction, meteorological monitoring, and so on. With the rapid development of deep-learning technology, many representative methods including pyramid scene parsing network (PSPNet) [1], Deeplab [2], criss-cross attention network (CCNet) [3], unified perceptual parsing network (UperNet) [4], and so on have been proposed and have achieved remarkable results in the field of semantic segmentation research.

The above methods usually adopt the paradigm of pretraining on the ImageNet dataset [5] and then fine-tuning on specific tasks. Recently, the development of foundation models represented by generative pre-trained transformer (GPT)-4 [6], vision transformer (ViT), and so on has brought disruptive technological changes to the fields of natural language processing (NLP) as well as computer vision (CV). For the semantic segmentation task, Meta AI took the lead in proposing the foundation model called segment anything model (SAM) [7]. By pretraining on the SA-1B dataset, combined with the prompt engineering in the form of points, boxes, masks, or text, SAM can achieve zero-shot generalization on multiple downstream tasks and show comparable performance to some supervised methods. After that, some other foundation models including segment everything in context (SegGPT) [8], segment everything everywhere all at once (SEEM) [9], and so on were proposed for segmentation tasks, which can also achieve good performance.

These foundation models mostly focus on the CV field. Compared to natural scene images, remote-sensing images have a wide coverage area, complex land types, and limited annotation data. There are some limitations in directly transferring them to the remote-sensing field, as shown in Fig. 1. The details are as follows.

1) *Category Information Missing:* Although SAM can segment anything, it cannot recognize the category of objects, as shown in Fig. 1(a). An important task of semantic segmentation is to provide category information of each pixel, to support various practical applications.
2) *Single Prompt Box Limitation:* Compared with natural images, the remote-sensing image usually has a wider coverage with multiple objects. The box prompt of SAM only supports a single box, which makes it difficult to realize prompt learning for multiple objects.
3) *Poor Performance on Synthetic Aperture Radar (SAR) Images:* Due to the domain differences of the training data, as shown in Fig. 1(b), the performance of SAM on SAR images is not so good and needs to be further explored.

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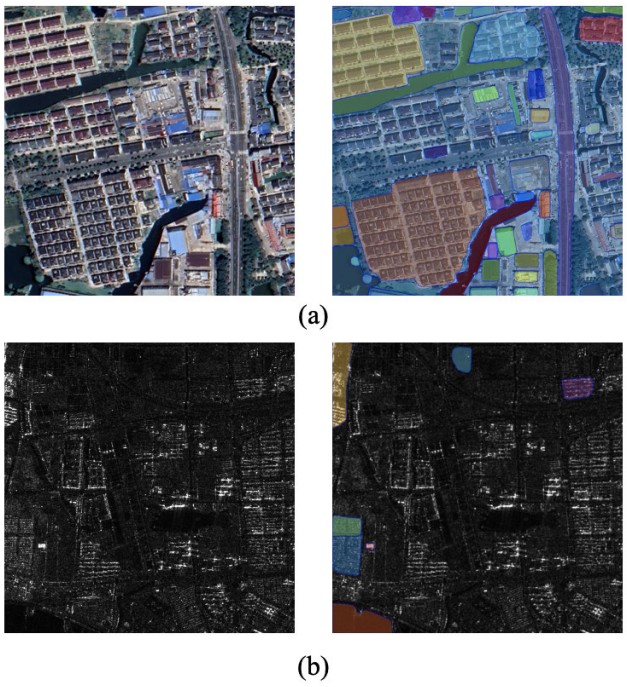


Fig. 1. Segmentation results of SAM on some remote-sensing images. (a) For optical remote-sensing images, the segmentation results of SAM are not fine enough under a complex background. Besides, there is no category information in the mask. (b) Poor performance on SAR images of SAM.

For the above problems, we propose a foundation model for multimodal remote-sensing image segmentation called RingMo-SAM in this article. To effectively train the model, a large-scale dataset containing millions of segmentation instances is constructed by collecting multiple open-source datasets in this field, including iSAID [10], ISPRS Vaihingen [11], ISPRS Potsdam [11], WHU-OPT-SAR [12], AIR-PolSAR-Seg [13], and so on. Considering the differences in the object categories in different datasets, we combed and merged the categories and finally formed the training set. Compared with natural images, the objects in remote-sensing images can be divided into two main types: instance-type (such as vehicles, aircraft, ships, etc.) and terrain-type (such as vegetation, land, river, etc.). Due to the difference in footprint, there is a significant category imbalance between these two types that will make model training challenging. To address this problem, as shown in Fig. 2, a category-decoupling mask decoder (CDMDecoder) is designed to achieve separate optimization for different types of objects. In addition, the prompt in SAM can be a set of points, a box or mask, free-form text, or, in general, any information indicating what to segment in an image. In this article, we try to embed the characteristic knowledge of multimodal remote sensing into prompt learning. Specifically, remote-sensing images are large in size and always contain multiple objects. Multibox prompt learning will be more suitable for these images. For SAR images, different objects have different scattering characteristics. Embedding the scattering information into the prompt encoder might be useful to improve model performance. Based on the above considerations, we build the RingMo-SAM framework.

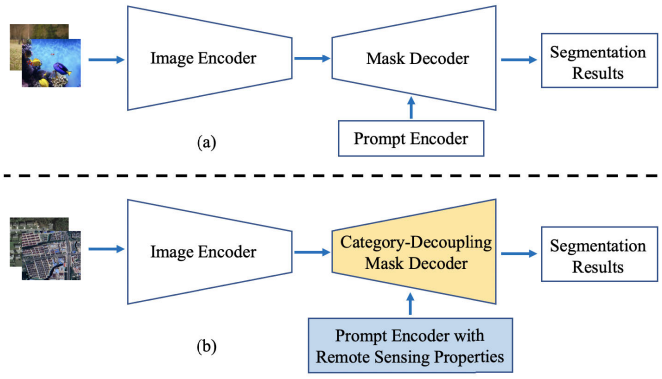


Fig. 2. Comparison of model frameworks between SAM and Ringmo-SAM. (a) Schematic of the SAM framework. (b) Schematic of the Ringmo-SAM framework.

In summary, the main contributions are as follows.

1) We propose a foundation model for multimodal remote-sensing image segmentation called RingMo-SAM, which can not only segment anything in optical and SAR remote-sensing data, but also identify object categories.
2) To solve the data imbalance problem between instance-type objects and terrain-type objects in remote-sensing images, a CDMDecoder is constructed to realize independent optimization for these two types of objects. In this way, the categorywise segmentation of various objects is achieved.
3) A prompt encoder embedded with the characteristics of multimodal remote-sensing data is designed. It not only supports multibox prompts to improve the segmentation accuracy of multiobjects in complicated remote-sensing scenes, but also supports SAR polarization decomposition prompts to improve the segmentation performance on SAR.

The rest of this article is organized as follows. Section II briefly introduces the representative methods of semantic segmentation, foundation models, and prompt learning. Then, we describe the proposed RingMo-SAM in Section III. Section IV presents comprehensive experimental results, which demonstrate the effectiveness of the method. Finally, the conclusion is outlined in Section V.

II. Related Works

A. Semantic Segmentation in Remote-Sensing Scenes

Deep-learning technology has brought great development to semantic segmentation. Many deep convolutional neural networks (DCNNs) are capable of adaptively extracting both shallow and deep features from images to achieve excellent performance on semantic segmentation tasks [14], [15], [16], [17], [18]. It allows the network to continuously learn various semantic concepts and develop a comprehensive understanding of complex scenes. With the development of the segmentation field, fully supervised learning approaches [1], [19], [20] have achieved significant success on the single-mode natural scene data represented by optics. Compared to natural scene data, remote-sensing images offer a unique imaging perspective,

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encompass diverse target categories, and exhibit minimal interclass differences. A network trained solely on feature representations from single-modal data falls short of meeting the demands of contemporary remote-sensing interpretation tasks. Researchers are beginning to introduce additional modal data information [21], [22], [23] to enrich feature representations and guide the network to learn more efficiently.

Due to the richer scene characterization information provided by multimodal data, significant progress has been achieved in multimodal remote-sensing semantic segmentation [24]. The early deep fusion methods focus on integrating multimodal features, for example, Kampffmeyer et al. [25] superimposed all the multimodal data to be input, then extracted the features, and completed the semantic segmentation of the urban scene according to the extracted features. Couprie et al. [26] utilized depth information, thereby enhancing the ability to distinguish objects with similar depth-based features and spatial characteristics. Audebert et al. [27] employed a fusion of LiDAR and multispectral data, along with a multiscale approach, to extract spatial contextual information and achieve precise building distribution in urban environments. Srivastava et al. [28] proposed a multimodal deep-learning approach to explore available image information for additional information clues and increase the comprehension of urban land use by integrating aerial images and ground images. Then, with the development of comparative learning, some advanced networks are introduced into the training of auxiliary multimodal data. Jain et al. [22] enhanced the multimodal pretraining of simple framework for contrastive learning of visual representation (SimCLR) [29] by generating positive pairs from diverse satellite collections, thereby preventing the model from solely focusing on local features.

However, most of the above methods are developed for some specific remote-sensing images with poor generalization. It is difficult to extend to the increasingly updated remote-sensing data scenarios to play a full role. Our goal is to propose a universal foundation model that accommodates multiple combinations of remote-sensing modalities in response to the trend toward large-scale models.

B. Foundation Models

In recent years, the NLP community has begun to focus on the research of large language models (LLMs), leading to a range of groundbreaking advancements, such as GPT-3 [30], GPT-4 [6], and chat GPT (ChatGPT) [31]. Accordingly, a revolution has begun in the CV field, with researchers actively exploring large vision models (LVMs) [32], [33], [34], [35]. The common characteristic of the above models is to extensively pretrain the model on a vast collection of datasets using a task that enables prompt learning [36] for some downstream tasks. The field of CV is shifting toward a new research trend focused on task-agnostic foundational models, which have strong generalization ability. Inspired by the above work, Sun et al. [37] proposed the first generative self-supervised foundation model framework of remote sensing called RingMo. It has reached state-of-the-art (SOTA) performance on multiple remote-sensing downstream tasks.

In the field of semantic segmentation, Meta AI has proposed the SAM [7] recently, which is an interactive and promptable segmentation foundation model. SAM has acquired a comprehensive understanding of an "object" as a fundamental concept in CV and demonstrated strong generalization and zero-shot performance. Around the same time as SAM, the Beijing Academy of Artificial Intelligence (BAAI) proposed the SegGPT [8], which unifies different segmentation tasks into a common learning framework and randomly assigns color maps to each data sample based on context. Besides, a more consummate segmentation system is proposed in SEEM [9] by introducing a wider range of prompts, such as points, boxes, and region-related prompts from another image.

Some researchers have explored and further developed models with more interactive features, such as open-vocabulary segmentation (OVSeg) [38] with open segmentation, and Grounding DETR with Improved denoising anchor boxes (DINO) [39] with segmentation, detection, and editing functions. These approaches have yielded remarkable results and offer new hope for the realization of general AI visual understanding. With the development of foundation models, the Adapter [40] concept is proposed within the NLP community and is initially developed as a tool for fine-tuning downstream tasks on large foundation models. A series of related Adapter algorithms are gradually proposed, which address dataset incompatibility issues and play a crucial role in transfer learning and downstream task adaptation training. Chen et al. [41] have successfully applied adapters to enhance the performance of SAM and achieved superior results in camouflage object detection and shadow detection, effectively extending the application of SAM in downstream tasks. These models are mainly concentrated in the CV field. The performance of foundation models in the field of remote-sensing remains to be further explored.

C. Prompt Learning

The previous DCNN methods usually adopt the paradigm of pretraining on the ImageNet and then fine-tuning on specific tasks. However, fine-tuning is prone to overfitting, and it is not flexible enough for different downstream tasks. Researchers in the field of NLP have pioneered the concept of prompt [36] to enhance the input text for better comprehension by pretrained language models. As a corresponding technique, prompt learning has evolved from discrete [30] to continuous [42], [43] and then fine-tuning large-scale pretrained models [44] to new tasks with improved performance and efficiency. Accordingly, recent work [45], [46] extends prompt tuning to the field of CV, which exhibits strong generalization to downstream tasks even in few-shot or zero-shot settings. Colorful prompt tuning (CPT) [47] presents a principled approach to jointly consider visual and textual semantics for cross-modal prompt configurations. Domain adaptation via prompt learning (DAPL) [48] facilitates the acquisition of distinct domain representations and customizes domain-specific contexts for each domain, thereby bridging the gap between multimodality approaches and domain adaptation techniques. Furthermore, some methods focus on constructing better prompting texts. The method proposed by Jiang et al. [49] can automatically generate diverse prompts and a method that can combine answers from different prompts.

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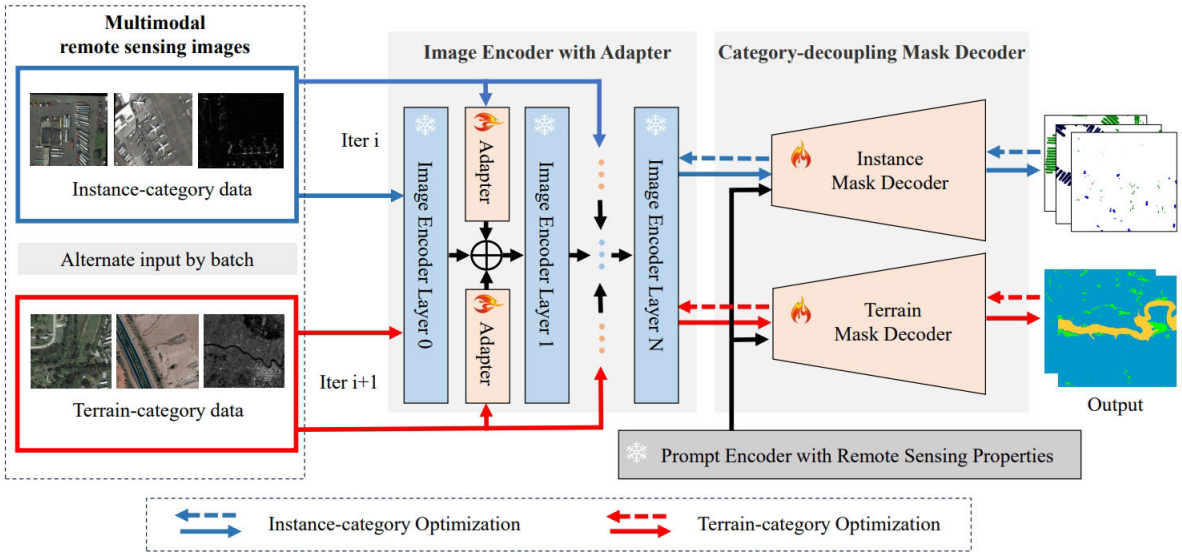


Fig. 3. Whole framework of our proposed RingMo-SAM. The image encoder with an adapter is used to output an image embedding. Then the CDMDecoder can decouple the SAM's mask decoder into an instance category decoder and a terrain category decoder so that they are optimized alternately according to batches during training, to ensure that the model fits adequately to both types of data. Besides, a prompt encoder embedded with the property of multimodal remote-sensing images is designed to improve the segmentation performance.

However, all of the aforementioned approaches have been exclusively applied to vision language models that are limited to input from text encoders. Some researchers are beginning to explore the universality and practicality of learning through purely visual prompting. Specifically, researchers utilize prompts to modify inputs and restructure downstream tasks in alignment with the initial pretraining task. Jia et al. [50] proposed the incorporation of task-specific prompts in the input space, which outperforms other fine-tuning protocols and reduces storage costs.

Bahng et al. [51] examined the influence of prompt design, downstream dataset, and output transformation on adaptation performance and demonstrated the significant advantages of the visual prompting on contrastive language-image pretraining (CLIP). Liang et al. [52] utilized learnable prompts to replace masked tokens, resulting in additional improvements beyond those achieved by a fully fine-tuned model. Chen et al. proposed the RSPrompter [53], which is a new prompt learning technique that enhances the capabilities of the SAM model for facilitating instance segmentation in remote-sensing imagery. Furthermore, prompt tuning has been widely adopted in open-vocabulary tasks, such as object detection [54] and semantic segmentation [55]. Inspired by the above work, we try to design a prompt encoder embedded with the characteristics of multimodal remote-sensing data to improve the segmentation performance of foundation models.

III. Methodology

A. Extensions on SAM to RingMo-SAM

SAM is mainly composed of three parts, namely an image encoder, a prompt encoder, and a mask decoder. The combination of these three components supports segmentation tasks at various prompts such as points, boxes, and masks. After training on a large-scale dataset SA-1B, SAM achieves better segmentation results. This ability can be attributed to the fact that SAM has mastered the basic concept of objects from a billion segmentation samples. However, for segmentation tasks in remote sensing, SAM has several shortcomings. The category information of objects is of great importance for practical applications. However, the outputs segmented by SAM do not have the category information. In addition, although SAM performs passable in some optical remote-sensing images, it is less effective in optical images with complex backgrounds and SAR images. Therefore, it is necessary to construct a foundation model of segmentation that supports multimodal remote-sensing images.

Combining the above analyses, we propose the RingMo-SAM. The whole framework is shown in Fig. 3. The task definition of RingMo-SAM is a promptable segmentation model for multimodal remote-sensed images, with interaction in the form of multibox prompts and polarization scatter prompts

$$\begin{aligned} E_{\text{img}} &= \Psi_{i\text{-enc}}(I) \\ E_{p\text{-boxes}} &= \Psi_{p\text{-enc}}\left([(b_i^{t-l}, b_i^{b-r})]\right) \\ E_{p\text{-scatter}} &= \Psi_{p\text{-enc}}\left(S_m, S_n, S_o\right) \\ M_{\text{multi-categ}} &= \Psi_{m\text{-dec}}\left(E_{\text{img}}, E_{p\text{-boxes}}, E_{p\text{-scatter}}\right) \end{aligned} \tag{1}$$

where $\Psi_{i\text{-enc}}(\cdot)$, $\Psi_{p\text{-enc}}(\cdot)$, and $\Psi_{m\text{-dec}}(\cdot)$ denote the image encoder, prompt encoder, and mask decoder, respectively. (b_i^{t-l}, b_i^{b-r}) indicates the top-left point and bottom-right point of the ith prompt box of multiprompt boxes. S_m, S_n, and S_o denote the different scattering components in the polarization decomposition of the SAR image. E_{img}, $E_{p\text{-boxes}}$, and $E_{p\text{-scatter}}$ indicate image embedding, sparse embedding of multiboxes prompt, and scatter prompt embedding, respectively. $M_{\text{multi-categ}}$ denotes multicategory masks of segmentation results.

For the mask decoder with categories $\Psi_{m\text{-dec}}(\cdot)$, we propose decoupling ideas and decoupled training strategy for the problem of ambiguity between categories and the problem of

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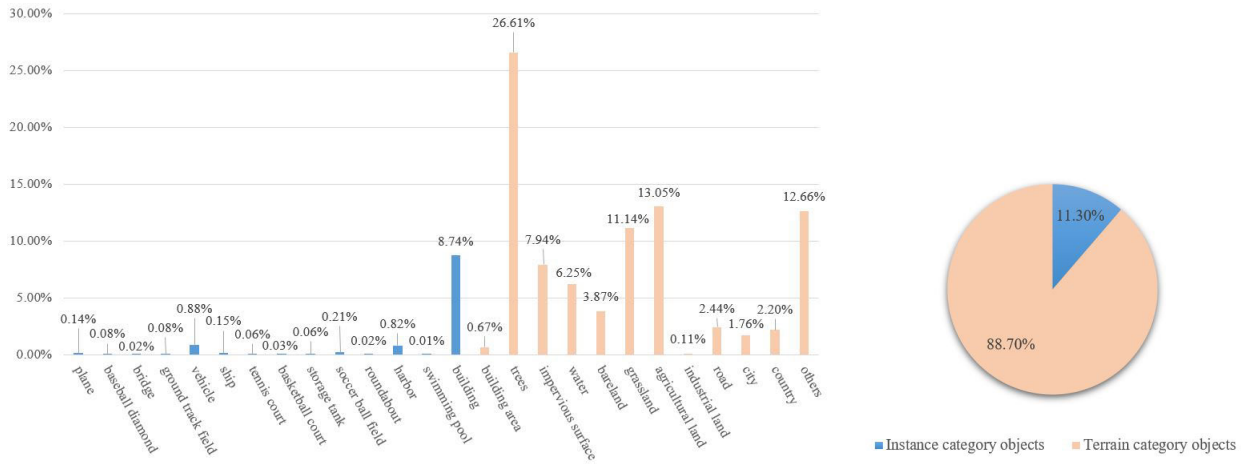


Fig. 4. Statistical distribution of categories in our collected remote-sensing segmentation datasets. Blue indicates the pixel proportion of the instance category and orange indicates the pixel proportion of the terrain category. There is a huge difference in the number of pixels between these two categories.

differences in the number of pixels in multimodal remote-sensing images, respectively, as detailed in Section III-B. As mentioned before, a prompt encoder embedded with the characteristics of multimodal remote-sensing data is of great interest for practical applications. We describe the design of multibox prompt $\Psi_{p\text{-enc}}([(b_i^{t-l}, b_i^{b-r})])$ in Section III-C1. The form of SAR polarization scatter prompt for SAR images $\Psi_{p\text{-enc}}(S_m, S_n, S_o)$ is detailed in Section III-C2.

B. Category-Decoupling Mask Decoder in RingMo-SAM

Compared to SAM, an important innovation of the RingMo-SAM is the ability of the model to predict category information. Our proposed approach works together to achieve it from three main perspectives: dataset construction, model structure modification, and training strategy adjustment. The construction of a remote-sensing image segmentation dataset with category annotations is the basis for achieving this capability, the modification of the model structure allows the model to have the potential to predict category information, and the adjustment of the training strategy optimizes the model's adaptation to the high number of categories, all three of which are critical for the model to achieve category prediction capability. Modifications to the model structure and adjustments to the training strategy are detailed in this section.

1) Structure of the CDMDecoder: To enable the model to learn object category information of remote-sensing images, we first collect a large amount of semantic segmentation data with category annotation. Different from natural scene images, we find that in multimodal remote-sensing images, there exists an obvious data imbalance problem between different categories of objects. Fig. 4 gives the statistical distribution of categories in our collected remote-sensing segmentation datasets. It can be seen that the pixel proportion of terrain categories is about 88.70%, while the pixel proportion of instance categories is only 11.30%. The single decoder structure in SAM will lead to the unbalanced learning optimization of models due to the large difference in the number of pixels, which is mainly manifested in the extremely poor segmentation of instance classes. To alleviate this problem,

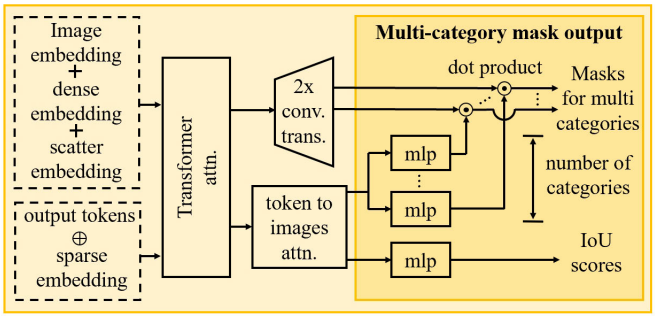


Fig. 5. Structural details of the instance and terrain mask decoder in the category-decoupling mask decoder. We implement the multicategory mask output with a parallel MLP structure.

we propose the CDMDecoder to decouple the optimization between instance categories and terrain categories.

The particulars of the decoupled model structure are shown in Fig. 3. We make structural improvements to SAM's lightweight mask decoder. It has the advantage of preserving the fast inference capability. First, we replicate one mask decoder into two mask decoders to resolve the problem of ambiguity between instance categories and terrain categories and then we modify the multilayer perceptron (MLP) network structure in the hypernetwork structure at the end of each mask decoder to parallel the multilayer structure for adapting to the output decoding structure for multicategory decoding, with the number of parallels of the MLP structure being determined by the number of categories of each mask decoder's output, as shown in Fig. 5.

2) Training Strategy of the CDMDecoder: Existing training strategy for segmentation task is holistic optimization, where all the data enter the network batch by batch for forward propagation and backward gradient updating, cycling through multiple epochs until the network converges. For RingMo-SAM, considering the pixel imbalance problem between the terrain category objects and the instance category objects, holistic optimization will lead to the problem of inadequate fitting of the instance category objects. Hence, we propose the decoupling training strategy as shown in Fig. 3. First, we organize the data into instance category data and terrain

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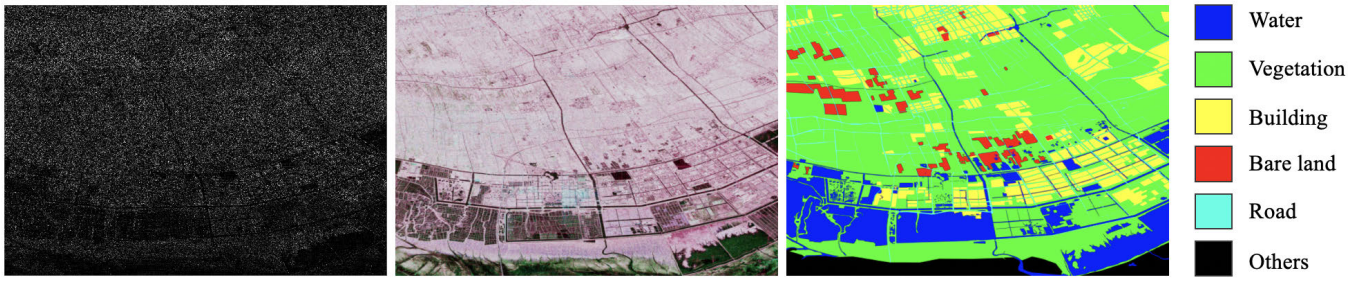


Fig. 6. Example of images of Zhejiang in the PolSAR-ZG dataset [56]. (a) SAR image. (b) Freeman–Durden decomposition pseudo-image. (c) Ground truth. In Freeman–Durden decomposition pseudo-image, each color represents a different polarized scattering component. Red denotes double-bounce scattering, green denotes volume scattering, and blue denotes surface scattering.

category data. Then, the two types of data are alternately fed into the model batch by batch in training for forward propagation and backward gradient updating. Only one type of data is input into each batch, that is, only instance category data or only terrain category data. The image encoder and the prompt encoder are shared, but each batch of data goes through only one decoder corresponding to the data type for forward propagation and backward gradient propagation during training. Such a training strategy avoids the issue of inadequate fitting bringing about the huge difference in the number of pixels between the terrain category data and the instance category data.

C. Prompt Learning in RingMo-SAM

The RingMo-SAM is a promptable segmentation model for multimodal remote-sensing images with two new promptable forms based on the properties of multimodal remote-sensing images: multibox prompt and SAR polarization scatter prompt.

1) Multibox Prompt: From the demo and implementation code shown by SAM, the form of the box prompt supported by SAM is a single box. It is not friendly to the practical application of remote-sensing images, which are characterized by the presence of numerous target objects [57]. Therefore, RingMo-SAM designs a multibox prompt form to eliminate this drawback.

We encode multiple prompt boxes of one input image together as a part of the sparse embedding to participate in the decoding process of the mask decoder, which in turn has a significant impact on optimizing the performance. The coding of multiple boxes is done one by one. Each box of multiboxes is coded by treating the top-left point and bottom-right point as two parallel dimensions, first normalized coding them, then coding them with Gaussian blurring, and finally adding additional embedding to the Gaussian blur-coded high-dimensional features. After multiple boxes are sequentially encoded as high-dimensional features, the corresponding multiple high-dimensional features are contacted together into a high-dimensional feature vector as part of the sparse embedding.

2) SAR Polarization Scatter Prompt: Different from optical remote-sensing images, SAR images are very different from human visual perception, and the information contained in them is difficult to understand. Therefore, SAR image segmentation has always been a very challenging subject. In PolSAR images, polarization information can be obtained by polarization decomposition, which can reveal the properties of objects. Fig. 6 gives the Freeman–Durden decomposition pseudo-image of the PolSAR-ZG dataset [56]. The specific decomposition components are obtained through the following equation:

$$\begin{aligned} <|\, S_{\mathrm{hh}} \,|^2> &= f_s \,|\, \beta \,|^2 + f_d \,|\, \alpha \,|^2 + f_v \\ <|\, S_{\mathrm{vv}} \,|^2> &= f_s + f_d + f_v \\ <|\, S_{\mathrm{hh}} S_{\mathrm{vv}}^* \,|^2> &= f_s \beta + f_d \alpha + f_v/3 \\ <|\, S_{\mathrm{hv}} \,|^2> &= f_v/3 \end{aligned} \tag{2}$$

where f_s, f_d, and f_v are the surface, double-bounce, and volume (or canopy) scatter contributions, respectively. S_{hh}, S_{hv}, and S_{vv} represent the elements of the radar backscatter image in HH polarization, HV polarization, and VV polarization, respectively. If $\mathrm{Re}(S_{\mathrm{hh}} S_{\mathrm{vv}}^*)$ is negative, the double-bounce scatter is dominant in the remainder, and $\beta = 1$. If $\mathrm{Re}(S_{\mathrm{hh}} S_{\mathrm{vv}}^*)$ is positive, the surface scatter is dominant and $\alpha = -1$.

Each color represents different polarization scattering components. Specifically, red denotes the double-bounce scattering, green represents the volume scattering, and blue expresses the surface scattering. It can be seen that the areas in the images of similar categories tend to show similar color components, while the areas of different categories show a large diversity of color components. Besides, different terrain categories usually exhibit different scattering properties. For example, water mainly present surface scattering, vegetation mainly present volume scattering, and man-made objects (such as buildings) mainly present double-bounce scattering. Besides, the polarization information obtained by the polarization decomposition methods can reveal the characteristics of the targets [58]. Therefore, we take the polarization feature in prompt engineering to improve the segmentation results of SAR images.

After obtaining the SAR polarization scatter results from the polarization decomposition, we design the prompt coding network for coded mapping of these SAR polarization scatter results to the corresponding SAR images. Specifically, first, we downsample the SAR polarization scatter results in spatial dimension to a quarter of the input size. After that, we perform downsampling with learnable parameters using a tandem of three modules: the 2-D convolution, the Gaussian error

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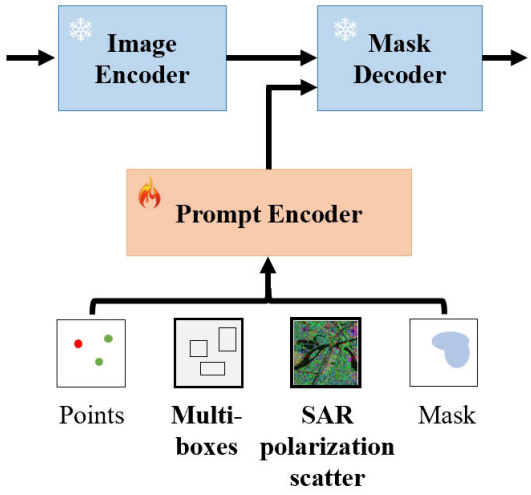


Fig. 7. Prompt encoder embedded with the properties of multimodal remote-sensing data in RingMo-SAM. It supports multibox prompts, which improves the segmentation accuracy of multiobjects in complicated remote-sensing scenes, as well as supports SAR characteristic prompts to improve the segmentation performance on SAR.

linear unit (GeLU) activation function, and the BatchNorm layer. Such downsampling is repeated twice, each time downsampling twice in the spatial dimension, and the vector obtained after two downsampling is reduced to one-sixteenth of the input image in the spatial dimension. Next, we use convolution to upscale the vector obtained from downsampling in the channel dimension to 256, named scatter embedding. The obtained high-dimensional vectors are consistent with the image embedding extracted by the image encoder in both channel dimension and spatial dimension. In the mask decoder, the scatter embedding is added to the image embedding on an element-by-element basis as part of the input, as shown in Fig. 5.

3) Prompt Encoder Refinement Training for RingMo-SAM: The encoding process in the multibox prompt form and SAR polarization scatter prompt has a small number of parameters that need to be updated during training. We adopt a fine-tuning approach to update these parameters. As shown in Fig. 7, we freeze the rest of the parameters such as the image encoder and the mask decoder except for multibox prompts and SAR polarization scatter prompt encoding and fine-tune only this part of the parameter in these two prompts.

IV. Experiments

To validate the performance of RingMo-SAM, we conducted extensive experiments on several multimodal remote-sensing datasets. In this section, we first introduce the restructured datasets, implementation details, and evaluation metrics. Then, the ablation experiments, comparative experiments, and some zero-shot experiments are reported with qualitative analysis.

A. Experimental Datasets

1) iSAID [10]: The iSAID is proposed for the instance segmentation task based on the DOTA dataset. There are 655 451 object instances in the iSAID dataset. The iSAID dataset contains 15 object categories, including swimming pool, bridge, tennis court, plane, small vehicle, soccer ball field, baseball diamond, basketball court, ship, ground track field, storage tank, harbor, large vehicle, helicopter, and roundabout. In this article, we use the predefined 1411 images for training and 458 images to validate our model.

2) ISPRS Vaihingen [11]: There are 33 remote-sensing images with three spectral bands and digital surface model (DSM) files in the Vaihingen dataset. The ground sampling distance (GSD) is 9 cm. The dataset contains five foreground classes including tree, low vegetation, car, building, and impervious surfaces, as well as one background class. We use the defined 16 images for training and 17 images for testing. Note that we do not use DSM in our experiments.

3) ISPRS Potsdam [11]: There are 38 high-resolution images with four spectral bands in this dataset. The spectral bands contain red, green, blue, and near-infrared. The GSD is 5 cm. The image size is 6000 × 6000 pixels. The semantic classes are the same as ISPRS Vaihingen. During the training, we used the defined 24 images and 14 images for testing.

4) WHU-OPT-SAR [12]: The WHU-OPT-SAR is a multimodal dataset. The data in this dataset contains two modalities: optical and SAR. It contains 100 pairs of multimodal images of 5556 × 3704 pixels in the same area. The optical images and the SAR images are collected from the Gaofen-1 satellite and Gaofen-3 satellite, respectively. The resolution is 2 and 5 m, respectively. There are seven categories in this dataset. In this article, 80% of the data is used for training and 20% for testing.

5) AIR-PolSAR-Seg [13]: The PolSAR image in this dataset is collected from the Gaofen-3 satellite. It has a resolution of 8 m and it is a full-polarimetric SAR sensor in the C-band. The size of the image is 9082 × 9805 pixels. The number of polarization modes in this dataset is four, including vertical–vertical (VV), horizontal–vertical (HV), horizontal–horizontal (HH), and vertical–horizontal (VH). The semantic classes are the same as in [13]. In this article, 80% of the data is used for training and 20% for testing.

6) AIR-Seg [59]: AIR-Seg was used for the 2020 Gaofen Challenge. The images in AIR-Seg are collected from the Gaofen-2 satellite with 0.8-m resolution. There are totally 17 categories of ground objects in this dataset, including road, building, shrub and tree, lawn, land, water body, impervious ground, roundabout, basketball court, football field, tennis court, baseball field, airplane, ship, bridge, vehicle, and others. The ratio of the training set and the test set is the same as the above dataset.

7) PolSAR-ZG [56]: PolSAR-ZG refers to PolSAR images covering the Zhejiang and Guangzhou regions in China. The data in PolSAR-ZG is collected from the Gaofen-3 satellite with 5-m resolution. The image size of Zhejiang is 6192 × 4888 pixels. The image size of Guangzhou is 5456 × 4708 pixels. There are six categories in PolSAR-ZG, including road, building, vegetation, water, bare land, and other areas.

B. Datasets Processing

To effectively train the Ringmo-SAM, a large-scale remote-sensing dataset containing millions of instances is constructed by processing the above multimodal datasets. The flow of data

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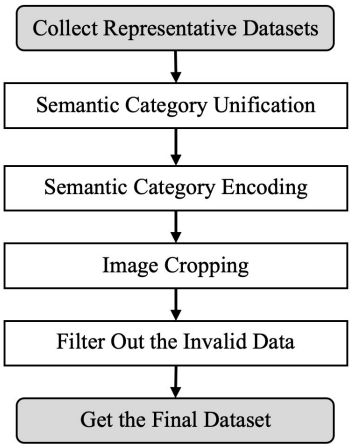


Fig. 8. Diagram of data processing.

TABLE I
CATEGORY INFORMATION USED IN THIS ARTICLE

Instance Category			
id	**class**	**id**	**class**
0	background	8	basketball court
1	plane	9	storage tank
2	baseball diamond	10	soccer ball field
3	bridge	11	roundabout
4	ground track field	12	harbor
5	vehicle	13	swimming pool
6	ship	14	building
7	tennis court		
Terrain Category			
id	**class**	**id**	**class**
0	building area	6	agricultural land
1	tree	7	industrial land
2	impervious surface	8	road
3	water	9	city
4	bare land	10	country
5	grass land	11	other

processing is shown in Fig. 8. First, we collect the representative datasets in the remote-sensing semantic segmentation field as much as possible. Considering the differences in the object categories in different datasets, we unify and encode the semantic categories of all collected data. There exists some semantic conflicts during this process. For example, the harbor in the iSAID dataset is marked as a bridge in the AIR-Seg dataset, as shown in Fig. 9(a). The ground track field in the iSAID dataset is labeled as the soccer ball field in the AIR-Seg dataset, as shown in Fig. 9(b). Besides, the building areas in SAR images are generally regionalized, but buildings in optical remote-sensing images are generally based on a roof, as shown in Fig. 10. We try to organize them uniformly and finally get 15 instance categories (including plane, ship, vehicle, etc.) and 12 terrain categories (including trees, water, bare land, etc.). The details are shown in Table I. Then, we crop the remote-sensing images and corresponding ground truth into (1024, 1024) patches for all data. Also, we filter out the invalid data in the dataset such as images with all backgrounds. In the end, the instance category samples contain 12 860 patches, and the terrain category samples have 16 081 patches.

TABLE II
ABLATION EXPERIMENTS ON THE iSAID DATASET. BEST IN **BOLD**

Baseline	CDMDecoder	Multi-box Prompt	mIoU
✓			3.89
✓	✓		57.91
✓	✓	✓	**57.99**

C. Implementation Details

To better preserve the knowledge of generality learned from SAM pretraining, we introduce the method of adapter, which adds a small number of parameters between the layers of the image encoder ViT contained in SAM. It is incorporated in all experiments in this article. Because the image encoder with adapter is not the main innovation of this article, there is not much introduction and experimental analysis. When training our RingMo-SAM, we freeze the image encoder and the prompt encoder and then update the parameters of the mask decoder. After that, the trained parameters are used to fine-tune the prompt encoder by freezing the image encoder and the mask decoder and updating the prompt encoder parameters. Training for all experiments in this article is done on four A100 GPUs. When training on large-scale remote-sensing data, the initial learning rate is 0.00016. The number of training epochs is 100, and the optimizer is Adam. When fine-tuning the prompt encoder, the initial learning rate is 0.0002, and the other settings are the same as the experimental settings above.

D. Evaluation Metrics

To evaluate the performance of the model, we use the mean intersection over union (mIoU) as the evaluation metrics, which are defined as follows:

$$\mathrm{mIoU} = \frac{1}{\mathrm{C}} \frac{\sum_{\mathrm{i}=1}^{\mathrm{C}} \mathrm{P_{ii}}}{\sum_{\mathrm{i}=1}^{\mathrm{C}} \sum_{\mathrm{j}=1}^{\mathrm{C}} \mathrm{P_{ij}} - \sum_{\mathrm{j}=1}^{\mathrm{C}} \mathrm{P_{ji}} - \mathrm{P_{ii}}} \tag{3}$$

where C represents the number of object categories, and P_{ij} is the numbers of pixels which belong to class i but are predicted as class j.

E. Ablation Experiments

To verify the effectiveness of the proposed methods, we carried out many ablation experiments on two representative multimodal datasets, with iSAID as an example of the optical remote-sensing dataset and PolSAR-ZG as an example of the SAR dataset.

1) Ablation Study on the iSAID Dataset: The iSAID dataset mainly contains instance categories. The results are shown in Table II. The baseline method uses a single decoder and feeds all remote-sensing images together into the model for

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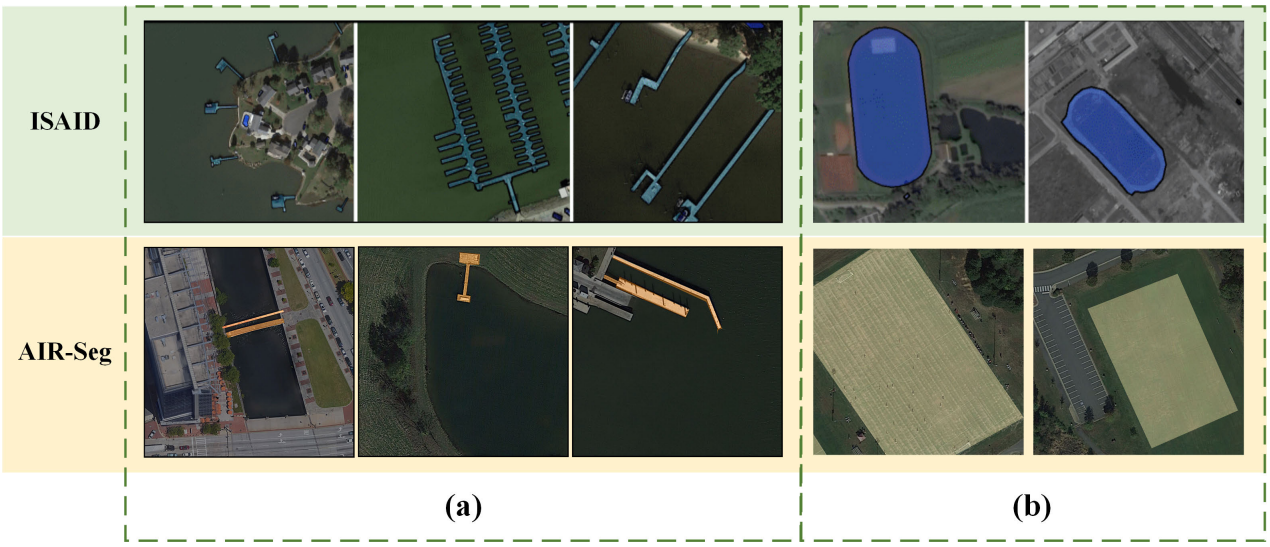


Fig. 9. Examples of some semantic category conflicts in different datasets. (a) Harbor in the iSAID dataset is marked as a bridge in the AIR-Seg dataset. (b) Ground track field in the iSAID dataset is labeled as a soccer ball field in the AIR-Seg dataset.

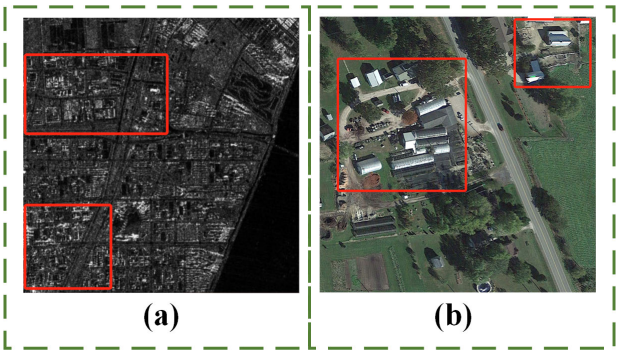


Fig. 10. Visualization of the building area with different remote-sensing modalities. (a) Building areas in SAR images are generally regionalized. (b) Buildings in optical remote-sensing images are generally based on a roof one by one.

training. The baseline method achieves a 3.89% mIoU on this dataset. Then, we changed the single decoder into our proposed CDMDecoder and used the training strategy of the CDMDecoder. Experimental results prove the effectiveness of our method, achieving a result of 57.91% mIoU. The CDMDecoder is beneficial for the model to learn the instance objects and the terrain objects separately. Finally, we add the multibox prompt, reaching a 57.99% mIoU. We argue that the multibox prompt is beneficial for the model to focus on the locations of multiple objects in remote-sensing images simultaneously. The experimental results have shown the effectiveness.

2) Ablation Study on the PolSAR-ZG Dataset: The PolSAR-ZG dataset contains full-polarization SAR images. The results on the PolSAR-ZG dataset are shown in Table III. The prediction of the baseline method on the PolSAR-ZG dataset achieves a 35.89% mIoU. When the model applies the CDMDecoder and corresponding training strategy, the model performance has increased by 4.34% mIoU. Based on the polarization scattering characteristics of SAR images, the SAR Polarization Scatter Prompt is designed. After adding the proposed prompt strategy, the model performance has

TABLE III
ABLATION EXPERIMENTS ON THE POLSAR-ZG DATASET. BEST IN **BOLD**

Baseline	CDMDecoder	Scatter Prompt	mIoU
✓			35.89
✓	✓		40.23
✓	✓	✓	**51.00**

further increased by 10.77% mIoU. It proves that the SAR polarization scatter prompt helps to refine the segmentation of the model for SAR images.

3) Parameter Adjustment Study for Learning Rate: We also explore the effect of different learning rates on the model with the SAR polarization scatter prompt. The experimental results are shown in Fig. 11. When the learning rate is $2e^{-}5$, the model achieves a 35.82% mIoU. When the learning rate increased to $8e^{-}5$, the model reached 40.95% mIoU. When the learning rate increased to $2e^{-}4$, the model performance was further improved with a 51.00% mIoU. We use this learning rate in our experiment.

F. Comparative Experiments on the SOTA Methods

1) Quantitative Analysis: We train the universal segmentation model RingMo-SAM for multimodal remote-sensing data based on all datasets and then test the model performance on several different datasets. For optical remote-sensing datasets, we test on the iSAID, ISPRS Vaihingen, and Potsdam datasets. For SAR images, we test on the representative PolSAR-ZG dataset. The experimental results on the iSAID dataset are shown in Table IV. The Nonlocal [63], Deeplab-v3 [19], and PSPNet [1] apply ResNet-50 [77] as the backbone. The UperNet [4] uses Swin-transformer-Base [78] as the backbone. According to the experimental results, it can be seen that the RingMo-SAM, which is a universal model, has achieved comparable accuracy to the specialized models, and even surpasses the SOTA methods in the categories of plane and storage tank.

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TABLE IV
PERFORMANCE COMPARISONS ON THE iSAID DATASET. BEST IN **BOLD**

Method	IoU per category(%)													mIoU
	PL	BD	BR	GTF	VE	SH	TC	BC	ST	SBF	RA	HA	SP	
U-Net [60]	37.35	36.5	7.5	5.5	42.75	49.0	78.6	22.9	0.0	9.7	46.5	45.6	38.0	32.30
FCN [61]	31.45	26.4	8.2	27.9	43.15	51.7	74.8	30.2	22.9	52.1	51.9	42.0	30.7	37.95
UperNet-Seco [62]	58.50	66.9	38.9	45.9	51.5	63.9	69.9	54.5	71.7	71.6	52.3	51.4	9.3	54.33
Non-local [63]	55.45	71.9	29.1	49.8	53.45	62.4	84.9	48.6	62.2	41.6	53.5	50.9	45.1	54.53
DANet [64]	55.65	71.4	28.6	52.5	49.8	60.2	84.7	50.9	63.0	63.3	40.6	48.8	46.1	55.04
CCNet [3]	54.00	72.4	30.5	52.2	48.75	56.3	85.3	49.2	63.2	67.1	51.8	48.7	43.6	55.62
Deeplab-v3 [19]	56.85	71.4	32.2	52.0	54.3	64.5	81.1	55.7	68.5	61.0	60.2	53.0	**47.1**	58.30
PSPNet [1]	58.05	**76.8**	31.2	53.4	52.75	63.9	86.5	56.7	68.3	67.4	52.6	53.0	44.0	58.82
SETR [65]	57.05	74.5	30.8	52.3	53.7	63.4	**86.9**	57.4	67.9	68.0	61.6	52.9	46.2	59.43
UperNet [4]	58.65	75.3	30.5	51.8	54.4	63.1	85.7	59.2	68.2	68.1	59.5	53.3	46.8	59.58
HRNet [66]	48.30	74	33.8	61.5	54.5	65.9	**86.9**	**59.4**	68.9	75.6	52.9	52.2	44.2	59.85
UperNet-ViTAEv2 [67]	60.30	72.2	**44.0**	**63.0**	**57.05**	**71.3**	70.4	57.4	**74.3**	**77.7**	**62.2**	**54.7**	47.0	**62.43**
SAM [7]	8.49	1.70	0.28	1.22	3.45	3.45	12.06	0.73	5.47	2.64	0.37	9.57	1.19	3.89
Ringmo-SAM	**76.67**	56.81	21.47	58.58	50.22	62.78	79.41	48.79	**74.3**	74.73	52.67	51.12	46.32	**57.99**

* plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), vehicle (VE), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP)

TABLE V
PERFORMANCE COMPARISONS ON THE ISPRS VAIHINGEN DATASET. BEST IN **BOLD**

Method	IoU per category(%)					mIoU
	Car	Building	Imp. Surf.	Low. Veg.	Tree	
Capsules-Unet [68]	1.30	44.22	52.88	19.43	54.92	34.55
SETR_PUP [65]	28.62	71.27	68.15	54.96	69.03	58.41
TNT [69]	26.49	73.28	69.15	54.83	69.04	58.56
PSPNet [1]	50.67	82.69	76.16	61.93	73.08	68.91
Transunet [70]	54.37	78.99	75.23	63.05	73.80	69.09
Deeplab-v3+ [20]	53.83	82.04	76.98	62.79	73.69	69.87
SAN [71]	55.98	82.53	76.89	62.88	73.42	70.34
BoTNet [72]	60.42	83.82	78.64	64.28	74.03	72.24
U-Net [60]	63.25	83.25	78.31	63.77	74.65	72.65
StransFuse [73]	62.79	84.27	78.97	**65.35**	**74.69**	73.21
DDCM-Net [74]	63.78	84.51	78.90	64.28	74.66	73.23
HCANet [75]	**64.17**	84.82	78.90	64.15	74.33	**73.27**
Ringmo-SAM	61.29	**89.10**	**79.70**	60.25	71.85	72.44

* Impervious Surface (Imp. Surf.); Low Vegetation (Low. Veg.)

The results prove the effectiveness of the proposed method on instance-categories segmentation.

We further compare the RingMo-SAM with existing methods on the ISPRS Vaihingen and Potsdam datasets. Tables V and VI show the results. The images in the Vaihingen and Potsdam datasets contain both instance objects and terrain objects. We output the final prediction result according to the output probabilities of the instance mask decoder and the terrain mask decoder. As for the Vaihingen dataset, the RingMo-SAM outperforms the superior models including PSPNet, Transunet, Deeplab-v3+, and so on. Specifically, it improves the mIoU by 3.53% over PSPNet, 3.35% over Transunet, and 2.57% over Deeplab-v3+. Besides, it achieves comparable accuracy to the hierarchical context aggregation network (HCANet), especially in categories of building and impervious surfaces. As for the Potsdam dataset, the RingMo-SAM gets similar results to the Vaihingen dataset. It achieves comparable accuracy to the U-Net and outperforms the superior models of Transunet by 4.12%, PSPNet by 2.71%, and Deeplab-v3+ by 2.16%.

We also experimented with the PolSAR-ZG dataset. As shown in Table VII, compared with optical remote-sensing image segmentation, the task of SAR image segmentation is more challenging. Objectively speaking, our RingMo-SAM, which is a multimodal universal model, still has a certain gap in segmentation performance compared to the SOTA specialized models on SAR data. But the results on certain categories including water, vegetation, and so on are still

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TABLE VI
PERFORMANCE COMPARISONS ON THE ISPRS POTSDAM DATASET. BEST IN **BOLD**

Method	IoU per category(%)					mIoU
	Car	Building	Imp. Surf.	Low. Veg.	Tree	
Capsules-Unet [68]	14.18	32.38	40.25	40.83	13.87	28.30
SETR_PUP [65]	57.73	72.38	67.27	55.25	47.69	60.06
TNT [69]	54.19	73.77	65.69	56.80	51.61	60.41
Transunet [70]	77.91	81.77	76.42	67.29	67.04	74.09
PSPNet [1]	73.90	86.11	78.66	68.72	70.12	75.50
Deeplab-v3+ [20]	77.31	85.79	78.90	68.57	69.69	76.05
SAN [71]	76.72	86.24	79.32	69.20	69.85	76.27
BoTNet [72]	78.06	87.21	80.10	68.98	69.00	76.67
StransFuse [73]	79.39	88.53	**81.41**	70.81	71.84	78.40
HCANet [75]	80.56	88.24	80.54	70.76	72.14	78.45
DDCM-Net [74]	80.23	88.48	81.35	70.75	72.16	78.60
U-Net [60]	**81.28**	87.30	80.94	**70.89**	**72.67**	**78.62**
Ringmo-SAM	80.77	**92.76**	77.56	70.01	69.94	78.21

TABLE VII
PERFORMANCE COMPARISONS ON THE PolSAR-ZG DATASET. BEST IN **BOLD**

Method	IoU per category(%)						mIoU
	Water	Vegetation	Bare land	Building	Road	Other	
FCN [61]	71.51	83.62	69.34	79.57	48.53	74.61	71.20
OCRNet [76]	75.07	85.12	71.89	83.50	53.46	76.46	74.25
UNet [60]	75.54	87.02	72.53	82.48	54.30	78.08	74.99
PSPNet [1]	76.64	86.88	73.60	84.14	56.38	78.91	76.09
Deeplab-v3+ [20]	**77.39**	**87.23**	**75.53**	**84.80**	**57.08**	**79.40**	**76.90**
Ringmo-SAM	76.21	62.97	8.05	67.75	25.41	65.62	51.00

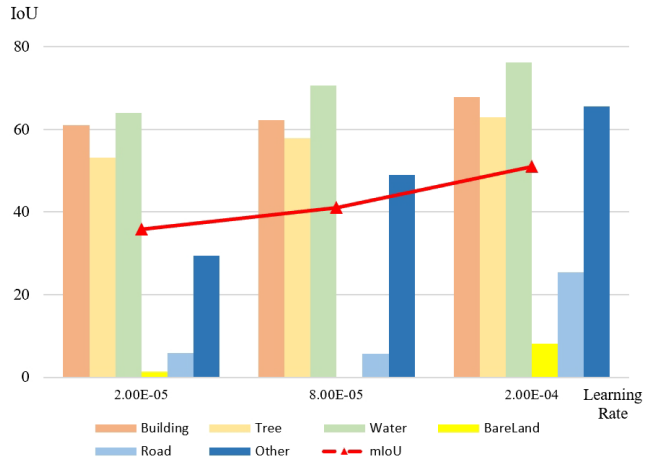


Fig. 11. Model performance with different learning rates in our experiment. When the learning rate is set to $2e^{-4}$, the model achieves a better performance with a 51.00% mIoU. We use this learning rate in our experiment.

competitive. There still remains some room for improvement in the SAR segmentation task.

2) Qualitative Analysis: To observe the segmentation performance of the Ringm-SAM, we perform the visualization comparison of the results. The visualization results on the iSAID dataset are shown in Fig. 12. We can see that the prediction segmentation results of RingMo-SAM are superior. Especially for the category of cars and baseball courts, the segmentation results of RingMo-SAM are more refined. While SAM tends to segment cars and baseball courts into contiguous regions, leading to a rougher mask. Also, the multibox prompt in our RingMo-SAM can help to focus on multiple objects in images at the same time. Thus, the masks predicted by our model are more complete in the segmentation of multiple objects, such as planes and storage tanks. While the masks predicted by SAM may miss some objects in the images.

We also give some visualization comparisons on the ISPRS Vaihingen and Potsdam datasets. Vaihingen and Potsdam datasets contain both instance categories and terrain categories. Among them, buildings and cars belong to the category of instance type. Low vegetation, trees, and impervious surfaces belong to the category of terrain type. The classification method is based on the output logits by the two decoders. Specifically, for each pixel, if the outputs of the two decoders are background and foreground, respectively, the final prediction results are the foreground category. If the output results of both decoders are foreground, we will concatenate the output logits according to the channel. Finally, we take the category with the highest predicted logits. The final prediction results in the two datasets are shown in Figs. 13 and 14. Our method

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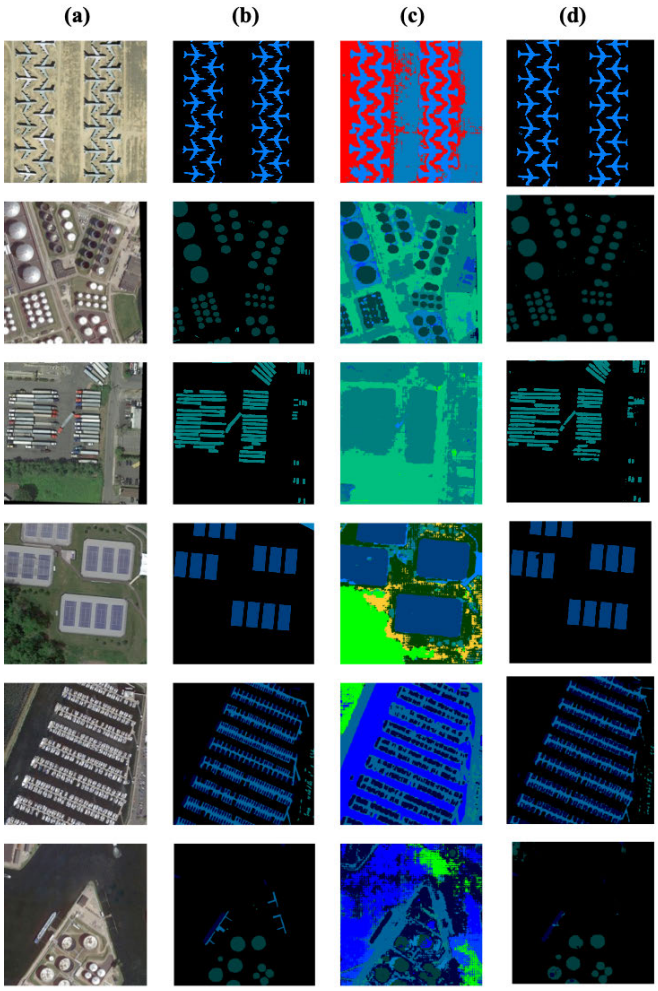


Fig. 12. Visualization of semantic segmentation results on the iSAID dataset. (a) Input images. (b) Ground truth. (c) Visualization of SAM. (d) Visualization of ours.

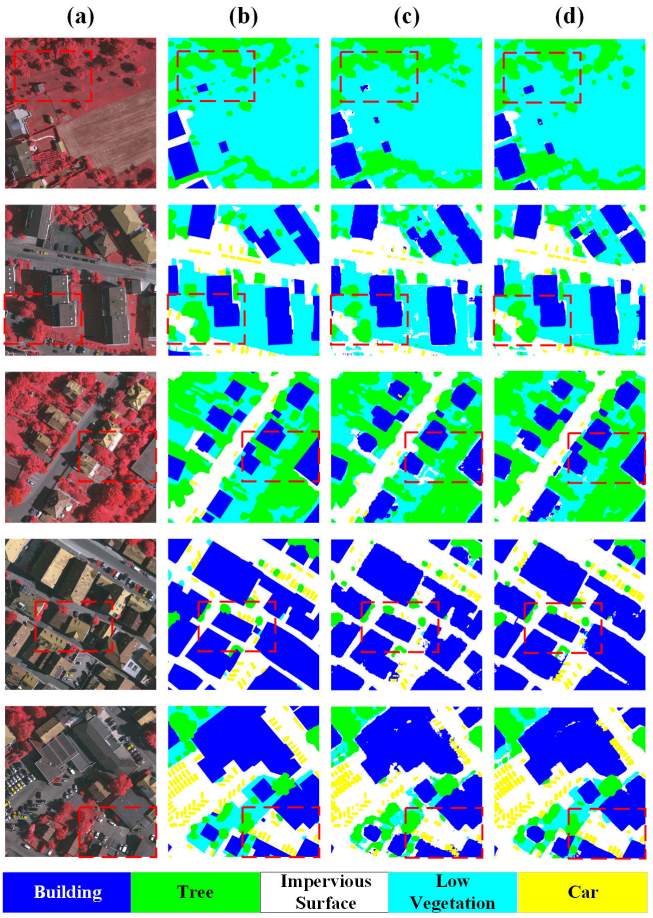


Fig. 13. Visualization of semantic segmentation results on the ISPRS Vaihingen dataset. (a) Input images. (b) Ground truth. (c) Visualization of SAM. (d) Visualization of ours.

has obvious advantages for the prediction of multiple semantic categories. Especially for small cars, whether in dense parking lots or on the road, our method predicts more accurately.

Furthermore, the qualitative results on the PolSAR-ZG dataset are shown in Fig. 15. We visual the segmentation mask of SAM and our RingMo-SAM. As can be seen from Fig. 15, the segmentation results of our model are closer to the ground truth. Especially for the category of road and vegetation, the segmentation results of RingMo-SAM are more accurate than the baseline methods. However, the segmentation boundaries of both models are relatively rough, remaining room for improvement in this field.

G. Zero-Shot Transfer Experiments

1) Zero-Shot Semantic Segmentation: To verify the zero-shot generalization ability of the foundation model, we perform zero-shot semantic segmentation experiments on the LoveDA dataset [79]. It is not used for training the RingMo-SAM model. Besides, the image resolution and scenes of this dataset are different from the training set, which is suitable for verifying the generalization performance of the model under different conditions. The LoveDA dataset contains 5987 images with 0.3-m resolution collected from Nanjing, Changzhou, and Wuhan in China. There are more than 166 768 instances in it. The categories contain both instance categories and terrain categories, including buildings, roads, water, bare land, forest, architecture land, and background. The visualization of segmentation results is shown in Fig. 16. Compared with SAM, our method has more accurate segmentation results for roads, water, trees, and so on. It is difficult for SAM to semantically label the segmented objects. SAM has a discontinuity in the prediction of objects, and the predictions of the foreground are also made for some pixels that are single or do not constitute objects. It can be seen that RingMo-SAM almost has the ability of zero-shot segmentation and can make more accurate predictions on different data distributions.

2) Zero-Shot Object Detection: We also perform the zero-shot object detection task on the FAIR1M [80] dataset. The FAIR1M dataset contains more than 15 000 images with a resolution greater than 1 m, including more than 1 million instance objects. The images in the FAIR1M contain multiple scenes such as cities, airports, and seaports. FAIR1M generally contains instance-type objects. We did not use this dataset for training. To verify the effectiveness of RingMo-SAM, we visualized some patches of size 1024 $\times$ 1024. The

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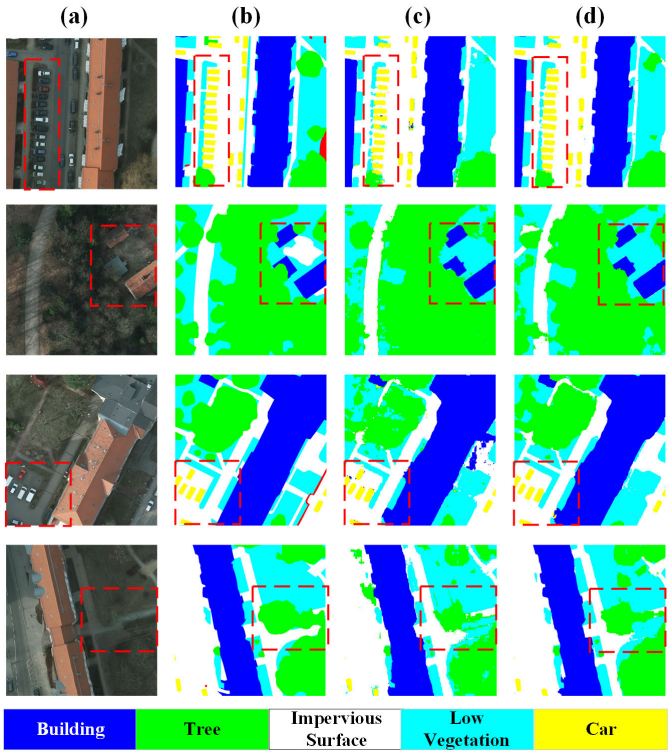


Fig. 14. Visualization of semantic segmentation results on the ISPRS Potsdam dataset. (a) Input images. (b) Ground truth. (c) Visualization of SAM. (d) Visualization of ours.

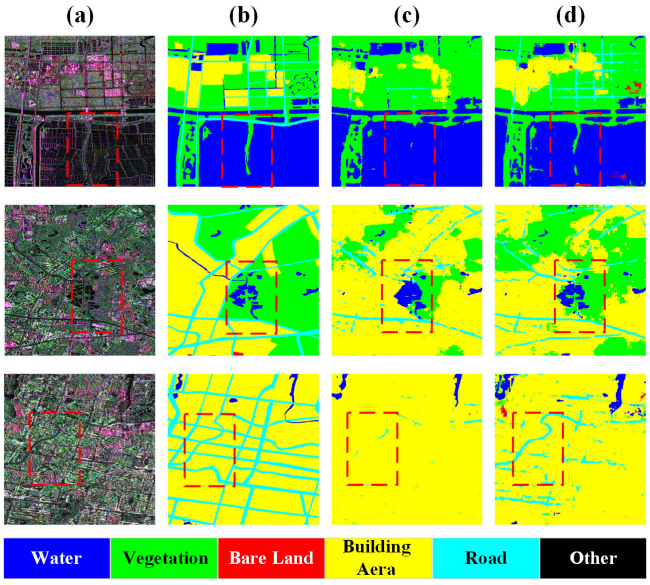


Fig. 15. Visualization of semantic segmentation results on the PolSAR-ZG dataset. (a) Input images. (b) Ground truth. (c) Visualization of SAM. (d) Visualization of ours.

visualization results are shown in Fig. 17. The first column represents the input images. The second column represents the result of object detection. The third column represents the result of zooming in on the object in Fig. 17(b). We can see that our model labels the positive frame of the object and the corresponding category information. The last column indicates the corresponding semantic segmentation mask. For instance,

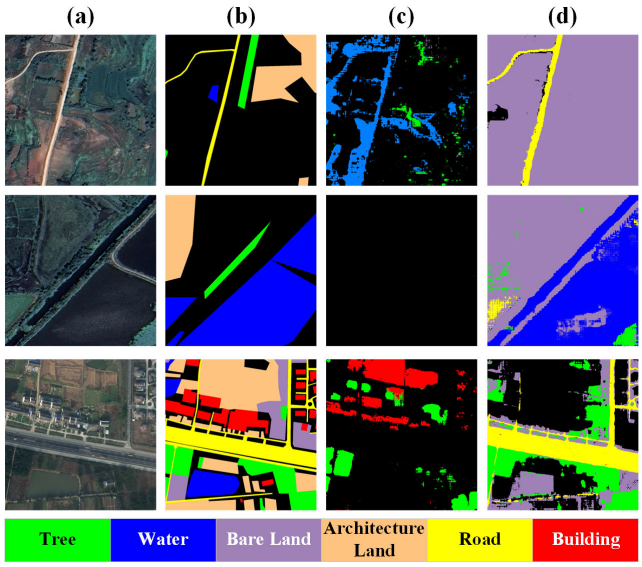


Fig. 16. Visualization of semantic segmentation results on LoveDA dataset. (a) Input images. (b) Ground truth. (c) Output of SAM. (d) Output of ours.

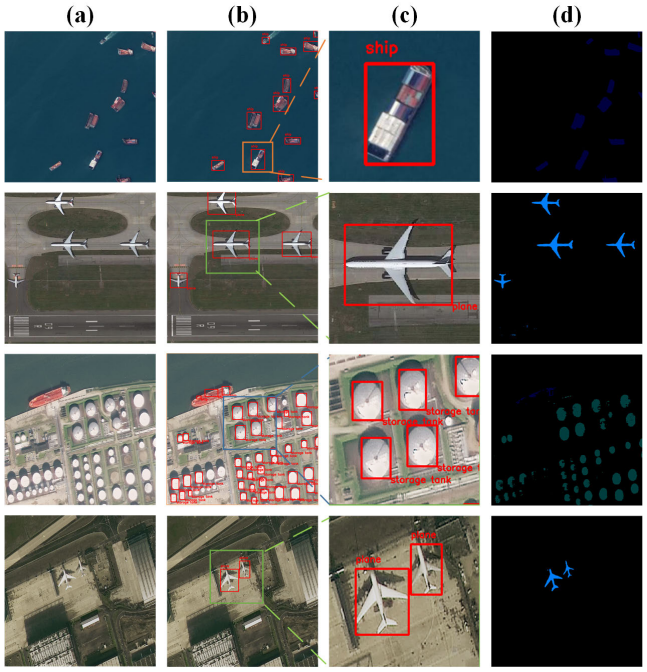


Fig. 17. Visualization of detection results on the FAIR1M dataset. (a) Input images. (b) Boxes of ours. (c) Output of zooming in the local area. (d) Mask of ours.

we see that RingMo-SAM makes an accurate prediction on object detection and semantic segmentation for unseen images.

H. Limitations

Although our RingMo-SAM has achieved good performance on some multimodal remote-sensing datasets, there are still some limitations to be further explored. The limitations mainly focus on the following two aspects. On the one hand, for dense small objects in the complex background of remote-sensing images, the segmentation mask is more likely to appear as a result of concatenation. This may be due to the

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lack of utilization of multiscale features by the RingMo-SAM, especially the shallow features, resulting in the loss of some detailed information during the downsampling process. On the other hand, the segmentation results of RingMo-SAM on SAR images still have a certain gap compared with the dedicated model. As we know, the imaging mechanism of the SAR image is complex, and the visual information in SAR images is very difficult to understand. It makes the segmentation of SAR images more challenging. It is necessary to further combine the properties of SAR images to further optimize the segmentation performance of RingMo-SAM.

V. Conclusion

In this article, we propose a foundation model for multimodal remote-sensing image segmentation which can not only segment anything in optical and SAR remote-sensing data but also identify object categories. By constructing an instance-type and terrain-type CDMDecoder, the categorywise segmentation of various objects is achieved. Besides, a prompt encoder embedded with the characteristics of remote-sensing data is designed. It not only supports multibox prompts to improve the segmentation accuracy of multiobjects in complicated remote-sensing scenes, but also supports SAR polarization decomposition prompts to improve the segmentation performance on SAR. Extensive experimental results have demonstrated the effectiveness of our method. In the future, we will explore how to better utilize remote-sensing characteristic knowledge as the prompter to further improve the performance of the model on multimodal remote-sensing data, especially in SAR image segmentation.

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